Unit: 6 Unit 7: Factoring
Topic: Expanding Special Patterns Perfect Square Trinomial

Name: ________________
Date: ____________

Homework (Jalapeno)

Q1. The ancient Egyptians built pyramids using $a^2 + 2ab + b^2$ blocks. What is the factored form of this expression?
(A) $(a + b)^2$
(B) $(a - b)^2$
(C) $(ab)^2$
(D) $a^2 + b^2$
(E) $2ab$

Q2. The Babylonians used the formula $x^2 - 2xy + y^2$ to calculate areas. What is the factored form of this expression?
(A) $(x + y)^2$
(B) $(x - y)^2$
(C) $(xy)^2$
(D) $x^2 + y^2$
(E) $2xy$

Q3. The ancient Greeks expanded the expression $(x+3)^2$. What is the result?
(A) $x^2 + 6x + 9$
(B) $x^2 - 6x + 9$
(C) $x^2 + 3x + 9$
(D) $x^2 - 3x + 9$
(E) $x^2 + 9$

Q4. The Mayans used the expression $a^2 - 2ab + b^2$ to calculate the area of a rectangle. What is the factored form of this expression?
(A) $(a + b)^2$
(B) $(a - b)^2$
(C) $(ab)^2$
(D) $a^2 + b^2$
(E) $2ab$

Q5. The Chinese expanded the expression $(x - 2)^2$. What is the result?
(A) $x^2 + 4x + 4$
(B) $x^2 - 4x + 4$
(C) $x^2 + 2x + 4$
(D) $x^2 - 2x + 4$
(E) $x^2 + 4$

Q6. The ancient Indians used the formula $x^2 + 2xy + y^2$ to calculate volumes. What is the factored form of this expression?
(A) $(x + y)^2$
(B) $(x - y)^2$
(C) $(xy)^2$
(D) $x^2 + y^2$
(E) $2xy$

Q7. The Phoenicians expanded the expression $(x + 5)^2$. What is the result?
(A) $x^2 + 10x + 25$
(B) $x^2 - 10x + 25$
(C) $x^2 + 5x + 25$
(D) $x^2 - 5x + 25$
(E) $x^2 + 25$

Q8. The ancient Sumerians used the expression $a^2 + 2ab + b^2$ to calculate areas. What is the factored form of this expression?
(A) $(a + b)^2$
(B) $(a - b)^2$
(C) $(ab)^2$
(D) $a^2 + b^2$
(E) $2ab$

Open Ended Questions (FRQ)

Q9. The ancient Egyptians built a pyramid with a base area of $x^2 + 4x + 4$. Factor the expression and find the dimensions of the base.

Q10. The Babylonians used the formula $x^2 - 6x + 9$ to calculate the area of a rectangle. Factor the expression and find the length and width of the rectangle.

Chef's Tips

- Tip 1: Check for perfect square trinomials
- Tip 2: Use the formula $(a + b)^2 = a^2 + 2ab + b^2$
- Tip 3: Use the formula $(a - b)^2 = a^2 - 2ab + b^2$